

What is Generative AI?

- Explain how it differs from traditional AI with at least two examples.

What is LangChain?

- Why is LangChain needed when building LLM applications? Explain its architecture and advantages.

What is Retrieval-Augmented Generation (RAG)?

- Explain the complete workflow of a RAG system with a diagram.

What is an API?

- Explain how an API works using a real-world example and describe the request-response cycle.

What is an API Key?

- Why is it required when using LLM APIs such as OpenAI? Explain the steps to generate and use an API key in Python.

Explain the Model Component in LangChain.

- What is the difference between an LLM and a Chat Model? Provide examples of each.

Explain the Prompt Component in LangChain.

- What is a `PromptTemplate`? Write a Python example demonstrating its use.

Explain the Chain Component in LangChain.

- How is a LangChain chain similar to a Machine Learning pipeline? Explain with an example.

What are Indexes in LangChain?

- Explain the roles of Document Loaders, Text Splitters, Embedding Models, Vector Stores, and Retrievers in building a RAG application.

Explain the Memory Component in LangChain.

- Compare the following memory types:
 - `ConversationBufferMemory`
 - `ConversationBufferWindowMemory`
 - `ConversationSummaryMemory`
 - Custom Memory